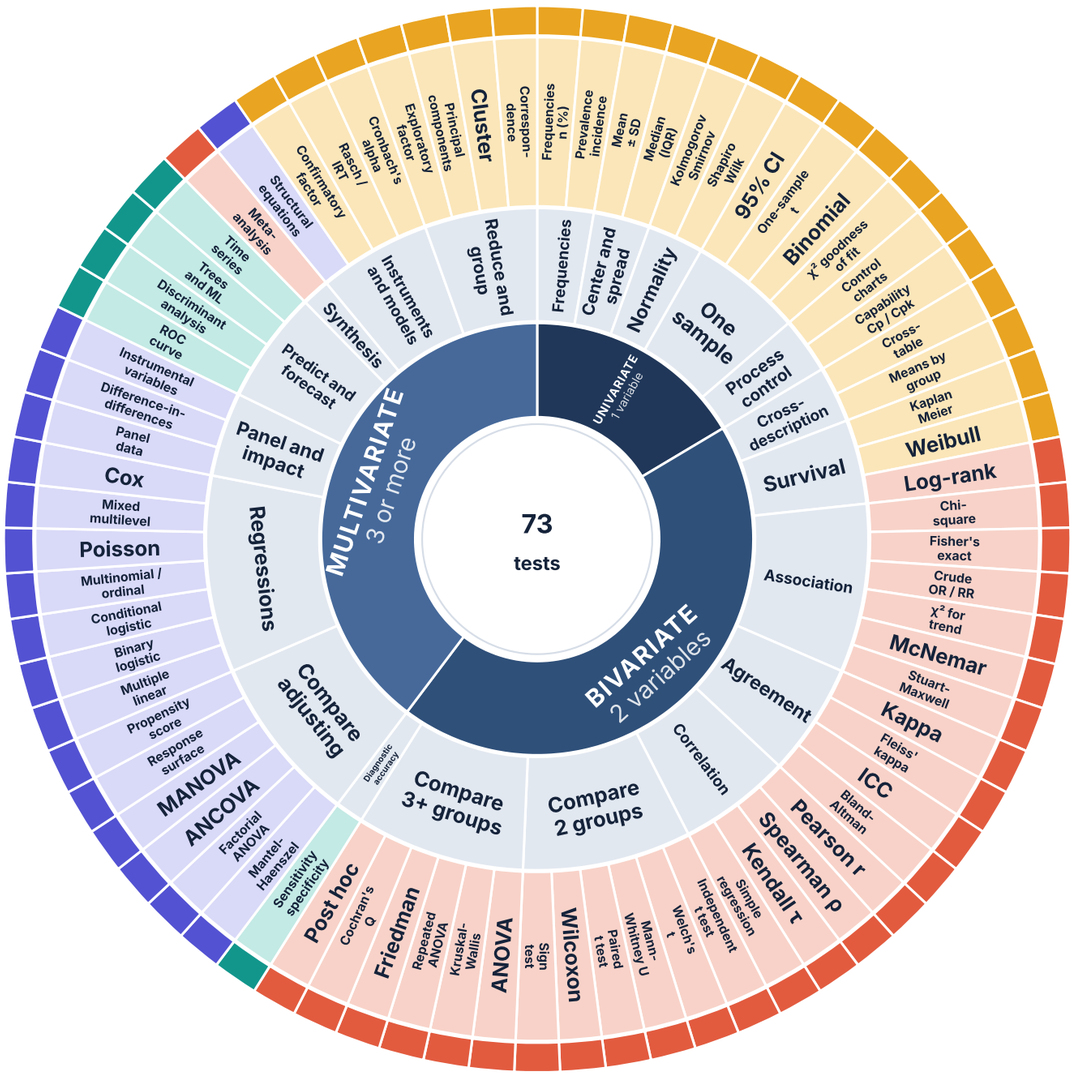
Statistics Wheel

Univariate, bivariate and multivariate tests by research scope: which verb the objective takes, which test fits and how the result is read.

Leidys-Helena Rivera-Quiroz
ORCID 0000-0003-1926-2521
Version 1.0 · 2026



Inner ring
How many variables are analyzed at once.

Middle ring
What you want to do with them.

Outer ring and color
The test and its scope.

SCOPES: EACH STEP INCLUDES THE PREVIOUS ONES (SHMUELI, 2010)

1 Descriptive

What is it like? How much is there?

Verbs: describe, characterize, estimate, quantify, determine the frequency, identify the frequency, examine, measure, validate an instrument, confirm the structure of a scale

Keywords: characteristics, profile, frequency, percentage, proportion, mean, median, prevalence, incidence, distribution, level, magnitude, reliability, validity, factor structure · **23 tests**

2 Relational

Are they associated? Do the groups differ?

Verbs: relate, associate, correlate, compare, contrast, establish the association, assess agreement

Keywords: relationship, association, correlation, difference, dependence, independence, agreement, between groups, before and after, crude OR, RR, r / rho, p-value, significance · **28 tests**

3 Explanatory

What is the effect of each factor when I control for the others?

Verbs: explain, estimate the effect, evaluate the effect, identify risk factors, determine associated factors (with adjustment), establish the determinants

Keywords: risk factors, associated factors, determinants, effect, adjusted, controlling for, confounding, covariates, adjusted OR, adjusted PR, HR, β coefficient, model · **17 tests**

4 Predictive

What will happen? How accurate is the test or model?

Verbs: predict, forecast, estimate the probability, classify, develop a model, validate a model, assess the accuracy

Keywords: prediction, forecast, probability, risk, predictive model, score, sensitivity, specificity, predictive value, ROC curve, AUC, calibration, validation, cutoff point · **5 tests**

A descriptive scope is not descriptive statistics. Descriptive statistics summarize the sample; inferential statistics generalize to the population with a 95% CI and p-value. Both can be uni-, bi- or multivariate, and the same measure (mean, r, OR) serves both.

How to read the p-value

- **p < .05:** reject H0; there is evidence of a difference or an association.
- **p ≥ .05:** do not reject H0; it does not show that the groups are equal (Altman & Bland, 1995).
- p does not give size: report the effect size and the 95% CI (Wasserstein & Lazar, 2016).
- p is not the probability that H0 is true (Wasserstein & Lazar, 2016).

Verbs that do not belong in a quantitative objective

~~to know (conocer)~~	→ describe, estimate, determine the frequency
~~to understand (comprender / entender)~~	→ explain (with an adjusted model) or relate
~~to interpret meanings~~	→ belongs to the qualitative approach
~~to inquire / to delve into~~	→ examine, estimate, compare
~~to unveil / to reflect on~~	→ have no measurable equivalent
~~to analyze (without saying what)~~	→ be specific: compare, associate, estimate the effect

Normal or not normal?

WHAT DO YOU WANT TO DO?	NORMAL DATA (PARAMETRIC)	NON-NORMAL OR ORDINAL DATA
Summarize a quantitative variable	Mean and standard deviation	Median and interquartile range
Compare 2 independent groups	Independent-samples Student's t test / Welch's t test	Mann-Whitney U test
Compare 2 related measurements	Paired-samples Student's t test	Wilcoxon signed-rank test / Sign test
Compare 3 or more independent groups	One-way ANOVA	Kruskal-Wallis test
Compare 3 or more related measurements	Repeated-measures ANOVA	Friedman test
Relate 2 quantitative variables	Pearson correlation	Spearman correlation / Kendall's tau
Compare the mean with a reference value	One-sample Student's t test	Wilcoxon signed-rank test

Shapiro-Wilk p < .05: the data are not normal; use the right-hand column (Shapiro & Wilk, 1965; Razali & Wah, 2011).

SIGNS OF INCOHERENCE

The objective says "to know" or "to understand" and the study is quantitative.
These are verbs of the qualitative approach or cannot be measured. Use **describe, estimate, compare, associate** or **estimate the effect**, depending on the scope.

The objective says "describe", but the results include p-values and comparisons.
The actual scope is **relational**. If the aim was only to describe, frequencies, means or medians are enough.

The objective says "determine the risk factors" and there is only a chi-square test.
Multivariable adjustment is missing. With chi-square the scope is **relational**; to be **explanatory** a regression is needed.

Shapiro-Wilk gave p < .05, but they report mean ± SD and Student's t.
With non-normal data, the right choice was the **median (IQR)** and the **Mann-Whitney U**.

They report the Mann-Whitney U as a "difference in means".
Mann-Whitney does not compare means. Medians are reported and the conclusion is that one group **tends to have higher values** (Fay & Proschan, 2010).

They conclude "no difference" or "the groups are equal" because p ≥ .05.
The correct conclusion is **no evidence of a difference was found** (Altman & Bland, 1995).

The objective says "predict", but they report no AUC, calibration or validation.
A **predictive** study must show how well the model performs on data that were not used to build it (Steyerberg, 2019).

They present a high Cronbach's alpha as proof that the scale measures a single dimension.
Alpha estimates reliability, not dimensionality. The structure is checked with **factor analysis**.

HOW TO CITE

Rivera-Quiroz, L.-H. (2026). *Rueda Estadística: Guía interactiva de pruebas estadísticas* [Statistics Wheel: An interactive guide to statistical tests] (Version 1.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23925941

CC BY 4.0 license. Concept and direction: the author; programming, drafting and simulation of examples with assistance from Claude (Anthropic). Cutoffs are conventions of the cited authors. Interactive version with 73 cards, examples by field, an assistant and practice mode: see the DOI.

POSTER REFERENCES (EACH TEST'S REFERENCES ARE IN THE INTERACTIVE VERSION)

Altman, D. G., & Bland, J. M. (1995). Absence of evidence is not evidence of absence. *BMJ, 311*(7003), 485. https://doi.org/10.1136/bmj.311.7003.485

Fay, M. P., & Proschan, M. A. (2010). Wilcoxon-Mann-Whitney or t-test? On assumptions for hypothesis tests and multiple interpretations of decision rules. *Statistics Surveys, 4*, 1–39. https://doi.org/10.1214/09-SS051

Razali, N. M., & Wah, Y. B. (2011). Power comparisons of Shapiro-Wilk, Kolmogorov-Smirnov, Lilliefors and Anderson-Darling tests. *Journal of Statistical Modeling and Analytics, 2*(1), 21–33.

Royston, P. (1992). Approximating the Shapiro-Wilk W-test for non-normality. *Statistics and Computing, 2*(3), 117–119. https://doi.org/10.1007/BF01891203

Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika, 52*(3–4), 591–611. https://doi.org/10.1093/biomet/52.3-4.591

Shmueli, G. (2010). To explain or to predict? *Statistical Science, 25*(3), 289–310. https://doi.org/10.1214/10-STS330

Steyerberg, E. W. (2019). *Clinical prediction models: A practical approach to development, validation, and updating* (2nd ed.). Springer. https://doi.org/10.1007/978-3-030-16399-0

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA's statement on p-values: Context, process, and purpose. *The American Statistician, 70*(2), 129–133. https://doi.org/10.1080/00031305.2016.1154108